YOUNGKI YOU

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PROFESSIONAL SUMMARY

Mechanism-driven structural and interaction biologist focused on how molecular recognition, protein interactions, and regulatory modifications control protein function across molecular and systems scales. Experienced in combining structural biology, XL-MS, LC-MS/MS proteomics, biochemical assays, and computational modeling to define mechanisms of protein interaction, conformational regulation, and functional organization in complex biological systems. Research trajectory spans protein sequence–structure analysis, mechanistic structural biology of membrane-associated signaling proteins, and systems-level proteomics of protein interaction networks and post-translational regulation, with emphasis on fundamental biological mechanisms relevant to disease and therapeutic discovery.

RESEARCH EXPERTISE

Mechanistic Structural Biology & Protein Regulation

- Structural and biochemical analysis of protein–ligand and protein-protein interactions in regulatory protein systems
- Ligand-dependent conformational regulation, membrane-associated signaling, and structure–function analysis
- X-ray crystallography, NMR-supported protein preparation, and integrative structural interpretation
- Protein engineering, construct design, mutagenesis, and stabilization for mechanistic studies
- Structure-guided analysis of interaction interfaces, dynamic protein states, and regulatory mechanisms

Protein Interaction Biology & Structural Proteomics

- XL-MS and LC-MS/MS workflows for mapping protein interaction architecture and complex organization
- Proteome-scale interaction mapping to identify interaction hubs and infer regulatory organization
- Structural interpretation of apolipoprotein, lipoprotein, biotin carboxylase, and microbial interaction systems
- Integration of biochemical, proteomic, and computational data to generate mechanistic models of protein assemblies
- Nanobody-based binder development for probing interaction specificity and complex organization

Mass Spectrometry, PTM Biology & Functional Proteomics

- LC-MS/MS-based proteomics for analysis of protein modifications and regulatory mechanisms
- PTM analysis including carbamylation, acetylation, phosphorylation, and methylation
- PTM crosstalk analysis connecting molecular modifications to functional and disease-relevant phenotypes
- Proteomics data analysis workflows using FragPipe, MaxQuant, Perseus, XlinkX, and pLink
- Familiarity with Thermo QE-HF, Lumos Tribrid, and Eclipse Tribrid platforms

Computational Structural Biology & Quantitative Analysis

- Molecular dynamics simulations and docking for characterization of protein–ligand and membrane-associated systems
- Structure prediction and modeling (AlphaFold, RoseTTAFold, IgFold)

- Structural visualization and refinement (PyMOL, Chimera, Discovery Studio)
- Python-, R-, and Linux-based workflows for data integration and quantitative analysis

Experimental Systems & Functional Characterization

- Protein expression, purification, and stabilization for structural and functional studies (affinity, ion exchange, SEC [ÄKTA], isotopic labeling)
- Biochemical and biophysical characterization of protein interactions and regulatory mechanisms
- Mammalian and bacterial cell culture, transformation, and transfection
- Immunoprecipitation, Western blotting, confocal microscopy, and functional characterization assays
- Experimental troubleshooting, project leadership, and cross-disciplinary collaboration

RESEARCH EXPERIENCE

Biological Sciences Division of Pacific Northwest National Laboratory Richland, WA

Postdoctoral Research Associate *02/2022 – 02/2026*

Research Focus: Mechanistic analysis of protein interaction networks and regulatory systems using structural biology, proteomics, and computational approaches.

- **Protein Interaction Architecture & Structural Proteomics**
  - **Defined interaction architectures of apolipoprotein and lipoprotein systems** by integrating biochemical assays, XL-MS, and computational modeling, revealing structural determinants of lipid-associated protein assemblies.
  - **Designed and characterized nanobody-based binders** to interrogate lipoprotein interactions, enabling quantitative analysis of binding specificity and structural organization of protein complexes.
  - **Applied XL-MS-derived constraints and structure prediction** to support mechanistic interpretation of protein assemblies and interaction interfaces.
  - **Characterized oligomeric assemblies of plant biotin carboxylase** through integrated cryo-EM, XL-MS, biochemical, and structural interpretation, providing insight into assembly-dependent functional regulation.
- **Proteome-Scale Interaction Mapping & Systems Biology**
  - **Mapped proteome-scale protein interaction networks** in *Chlamydomonas reinhardtii* and *Prochlorococcus* MED4, identifying interaction hubs and inferring functional organization across biological systems.
  - **Developed and applied XL-MS workflows for large-scale protein interaction analysis**, generating approximately 2,700 inter-protein crosslinks in cyanobacterial systems and enabling structural interpretation of prokaryotic interaction networks.
  - **Established reproducible experimental and analytical workflows** to improve robustness of complex protein interaction and structural proteomics datasets.
- **PTM Biology & Disease-Relevant Regulatory Mechanisms**
  - **Developed LC-MS/MS workflows** to profile carbamylation and PTM crosstalk, identifying more than 8,000 carbamylated and more than 7,000 acetylated peptides and expanding coverage of modification-dependent regulation.
  - **Defined PTM-dependent regulatory mechanisms** associated with type 1 diabetes and inflammatory signaling, including ARH3-mediated modulation of protein function.

- **Linked PTM landscapes to functional outcomes and disease-relevant phenotypes** through quantitative proteomics and mechanistic interpretation.
- **Revealed evolutionarily conserved methylation events** in bacterial systems through integration of large-scale proteomics datasets with orthogonal experimental validation.

- **Data Integration & Mechanistic Interpretation**
  - **Developed PTMOverlay**, a computational tool for visualization and interpretation of complex PTM datasets across biological systems.
  - **Integrated structural, biochemical, proteomic, and computational datasets** to generate mechanistic models of protein interactions, regulatory assemblies, and modification-dependent protein function.
  - **Collaborated across multidisciplinary teams** to connect unbiased proteomic discovery with targeted biochemical, structural, and functional studies.

The College of Pharmacy of University of Houston Houston, TX

Postdoctoral Fellow *01/2021 – 07/2021*

Graduate Research Assistant *08/2015 – 12/2020*

Research Focus: Structural and computational analysis of molecular recognition and conformational regulation in membrane-associated signaling proteins, centered on the Munc13-1 C1 domain.

- **Munc13-1 Structural Mechanisms of Ligand Binding and Membrane Interaction**
  - **Defined ligand-dependent conformational states of the Munc13-1 C1 domain**, linking molecular recognition to membrane-associated signaling and protein activation mechanisms.
  - **Determined structural determinants governing ligand binding and membrane interaction** through X-ray crystallography, biochemical characterization, molecular dynamics simulations, and docking.
  - **Determined apo and ligand-bound structural states of Munc13-1 C1**, defining binding modes and conformational changes associated with protein activation.
  - **Identified key residues, including Trp-588,** that regulate ligand binding, membrane association, and structure–function relationships.
  - **Engineered and characterized wild-type and mutant variants** to evaluate how structural perturbations alter protein function and regulatory behavior.
- **Protein Biochemistry, Cellular Function & Dynamic Regulation**
  - **Expressed and purified unlabeled and ^{15}N-labeled Munc13-1 C1 domain**, optimizing stability and yield for structural and biophysical studies.
  - **Used confocal microscopy** to monitor ligand- and stimulus-dependent translocation in cells, connecting molecular recognition to membrane-associated functional behavior.
  - **Investigated effects of small molecules,** including DAG-lactones, curcumin analogs, resveratrol, and ethanol to define how ligand binding modulates protein activation and membrane interaction dynamics.
- **Computational Structural Biology & Collaborative Modeling**
  - **Applied molecular dynamics simulations and docking** to characterize protein–ligand interactions across Munc13-1, PKC domains, proteasome targets, ACE2, and TET2–DNA systems.
  - **Collaborated on structure-based modeling studies** supporting target characterization, mechanism-of-action analysis, and hypothesis generation.

- **Developed standardized molecular dynamics workflows** that improved reproducibility and adoption within the laboratory.

Handong Institute of Life Science, Handong Global University Pohang, South Korea
Graduate Research Assistant *02/2010 – 09/2014*
Research Assistant *12/2006 – 02/2007; 07/2007 – 01/2010*
Research Focus: Computational analysis of protein sequence–structure relationships and structure-based functional modeling.

- Developed an amino acid substitution matrix using structural and physicochemical properties of protein blocks through PCA-based computational analysis.
- Investigated protein sequence–structure relationships underlying functional organization and evolutionary conservation.
- Applied computational structural approaches to evaluate antiplatelet compounds and intranasal vaccine candidates through structure-based analysis.

EDUCATION

Ph.D., Pharmacology and Medicinal Chemistry - University of Houston**,** Houston, TX, 2020
M.Sc., Life Science - Graduate School, Handong Global University**,** Pohang, South Korea, 2012
B.Sc., Life Science (minor in Computer Science) - Handong Global University, Pohang, South Korea, 2010

PUBLICATIONS

22 peer-reviewed publications (9 first-author)

Peer-Reviewed Publications

1. Powell, S.M., **You, Y.**, Ogbu, C., and Melchior, J.T. “The structural and functional journey of Apolipoprotein A-I through the human body.” *Current Atherosclerosis Reports* (2026): 38.
2. Madison, H.J., Dunn, L., **You, Y.**, Lemes Jorge, G., Paša-Tolić, L., Thelen, J.J., Van Doren, S.R., and Yokom, A.L. “Oligomeric assemblies of plant biotin carboxylase revealed by cryo-EM and cross-linking.” *Biochemical Journal* (2026): BCJ20250372.
3. **You, Y.**, Sarkar, S., Deiter, C., Elliott, E., Nicora, C., Mirmira, R., Sussel, L., and Nakayasu, E.S. “Reduction of chemokine CXCL9 expression by omega-3 fatty acids via ADP-ribosylhydrolase ARH3 in MIN6 insulin-producing cells.” *Proteomics* (2025): e202400053.
4. **You, Y.**, Many, G., and Nakayasu, E.S. “Protein carbamylation and proteomics: from artifacts to elucidation of biological functions.” *Frontiers in Analytical Science* (2025): 1512573.
5. Dakup, P.P., Diaz Ludovico, I., **You, Y.**, Rao, C., Flores, J., Bramer, L.M., Rewers, M., Webb-Robertson, B.J.M., Metz, T.O., Mirmira, R.G., Sims, E.K., and Nakayasu, E.S. “Challenges and opportunities in state-of-the-art proteomics analysis for biomarker development from plasma extracellular vesicles.” *Proteomics* (2025): e70036.

6. Sarkar, S., Morris, J., **You, Y.**, Sexmith, H., Street, S.E., Thibert, S.M., Attah, I.K., Hutchinson Bunch, C.M., Novikova, I.V., Evans, J.E., Shah, A.S., Gordon, S.M., Segrest, J.P., Bornfeldt, K.E., Vaisar, T., Heinecke, J.W., Davidson, W.S., and Melchior, J.T. "APOA2 increases cholesterol efflux capacity to plasma HDL by displacing the C-terminus of resident APOA1." *Journal of Lipid Research* (2024): 65(12), 100686.
7. Sarkar, S., Zheng, X., Clair, G., Kwon, Y., **You, Y.**, Swensen, A.C., Webb-Robertson, B.J.M., Nakayasu, E.S., Qian, W.-J., and Metz, T.O. "Exploring new frontiers in type 1 diabetes through advanced mass spectrometry-based molecular measurements." *Trends in Molecular Medicine* (2024).
8. Walukiewicz, H., Farris, Y., Burnet, M., Feid, S., **You, Y.**, Kim, H., Bank, T., Christensen, D., Payne, S.H., Wolfe, A., Rao, C., and Nakayasu, E.S. "Regulation of bacterial stringent response by an evolutionarily conserved ribosomal protein L11 methylation." *mBio* (2024): e01773-24.
9. **You, Y.**[†], Tsai, C.-F.[†], Patel, R., Sarkar, S., Clair, G., Zhou, M., Liu, T., Metz, T.O., Das, C., and Nakayasu, E.S. "Analysis of a macrophage carbamylated proteome reveals a function in post-translational modification crosstalk." *Cell Communication and Signaling* (2023): 21, 241.
   [†] These authors contributed equally.
10. **You, Y.**, Mathukumalli, K., and Das, J. "Comparison of the ligand binding site of C1 domains: a molecular dynamics simulation study of the C1 domain–phorbol 13-acetate–membrane system." *Biomolecular Structure and Dynamics* (2023): 11796–11809.
11. Vallejo, M.C., Sarkar, S., Elliott, E.C., Henry, H.R., Powell, S.M., Diaz Ludovico, I., **You, Y.**, Huang, F., Payne, S.H., Ramanadham, S., Sims, E.K., Metz, T.O., Mirmira, R.G., and Nakayasu, E.S. "A proteomic meta-analysis refinement of plasma extracellular vesicles." *Scientific Data* (2023): 837.
12. Das, J.[†], **You, Y.**[†], Mathukumalli, K., Ann, J., Lee, J., and Marquez, V.E. "Activation of Munc13-1 by Diacylglycerol (DAG)-Lactones." *Biochemistry* (2023): 2717–2726.
   [†] These authors contributed equally.
13. **You, Y.**, and Das, J. "Molecular dynamics simulation studies on binding of activator and inhibitor to Munc13-1 C1 in the presence of membrane." *Biomolecular Structure and Dynamics* (2022): 14160–14175.
14. **You, Y.**, Katti, S., Yu, B., Igumenova, T.I., and Das, J. "Probing the diacylglycerol binding site of presynaptic Munc13-1." *Biochemistry* (2021): 60(16), 1286–1298.
15. Khan, T.K., **You, Y.**, Nelson, T.J., Kundu, S., Pramanik, S., and Das, J. "Modulation of proteasome activity by curcumin and didemethylcurcumin." *Biomolecular Structure and Dynamics* (2021): 8332–8339.
16. **You, Y.**, and Das, J. "Effect of ethanol on Munc13-1 C1 in membrane: a molecular dynamics simulation study." *Alcoholism: Clinical and Experimental Research* (2020): 44(7), 1344–1355.
17. Blanco, F.A., Czikora, A., Kedei, N., **You, Y.**, Mitchell, G.A., Pany, S., Ghosh, A., Blumberg, P.M., and Das, J.

"Munc13 is a molecular target of bryostatin 1." *Biochemistry* (2019): 58(27), 3016–3030.

18. Czikora, A., Pany, S., **You, Y.**, Saini, A.S., Lewin, N.E., Mitchell, G.A., Abramovitz, A., Kedei, N., Blumberg, P.M., and Das, J. "Structural determinants of phorbol ester binding activity of the C1a and C1b domains of protein kinase C theta." *Biochimica et Biophysica Acta - Biomembranes* (2018): 1860(5), 1046–1056.
19. Das, J., Kedei, N., Kelsey, J.S., **You, Y.**, Pany, S., Mitchell, G.A., Lewin, N.E., and Blumberg, P.M. "Critical role of Trp-588 of presynaptic Munc13-1 for ligand binding and membrane translocation." *Biochemistry* (2018): 57(5), 732–741.
20. Pany, S., Ghosh, A., You, Y., Nguyen, N., and Das, J. "Resveratrol inhibits phorbol ester-induced membrane translocation of presynaptic Munc13-1." *Biochimica et Biophysica Acta - General Subjects* (2017): 1861(11), 2640–2651.
21. Pany, S., **You, Y.**, and Das, J. "Curcumin inhibits protein kinase Cα activity by binding to its C1 domain." *Biochemistry* (2016): 55(45), 6327–6336.
22. **You, Y.**, Jang, I.H., Lee, K., Kim, H.J., and Lee, K.H. "An approach for a substitution matrix based on protein blocks and physicochemical properties of amino acids through PCA." *Interdisciplinary Bio Central* (2014): 6(1), 3.

Manuscripts and Preprints

23. Neupane, N.P.[†], Mathews, J.T.[†], Lingaraju, S.[†], **You, Y.**[†], Mathukumalli, K., and Das, J. "Mitragynine and morphine stabilize different conformations of the μ-opioid receptor as revealed by molecular dynamics simulations." *ACS Omega* (under review).

    [†] These authors contributed equally.
24. Sarkar, S., Li, F., **You, Y.**, Elliott, E.C., Zemaitis, K.J., Kim, H., Markillie, L.M., Traeger, J., Powell, S.M., *et al.* "Insulin post-transcriptional regulation via PARP12-mediated ADP-ribosylation." *Cell Communication and Signaling* (under revision).
25. Krieger, C.[†], Everton, Z.[†], **You, Y.**[†], Lewis, B., Bank, T., Burnet, M.C., *et al.* "PTMOverlay: A proteomic tool to visualize post-translational modifications across evolution." *Proteomics* (under review).

    [†] These authors contributed equally.
26. Sarkar, S., Flores, J.E., Nune, N., Ramos-Rodriguez, M., *et al.*, **You, Y.**, and Nakayasu, E.S. "Multi-omics data compendium of pancreatic islet and β-cell responses to pro-inflammatory cytokines." *Scientific Data* (under review).
27. Van Fossen, E., Stephenson, M., Frank, A., Akella, S., Wooldridge, R., Wilson, A., **You, Y.**, Nakayasu, E.S., Egbert, R., and Elmore, J. "Efficient genetic code expansion tools enable in vivo study of lysine acetylation in non-model bacteria." *bioRxiv* (2025).

AWARDS

- 1st Place, Graduate Student Oral Presentation, College of Pharmacy of UH: Aug. 9, 2019
- Travel Award, The Biophysical Society: Mar. 2-6, 2019
- Community & Leadership Trainings Awards, Handong University: Mar.14, 2007

REFERENCES

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